

Influence of Nd(II) and Nd(III) Ions on the Speciation and Kinetic Dynamics of Radiolytic Transients in Molten LiCl-KCl Eutectic Salt Mixtures

Stephanie Castro Baldivieso,* Gregory P. Horne, Davis Bryars, Alejandro Ramos Ballesteros, Andrew R. Cook, Bobby Layne, Kazuhiro Iwamatsu, and Jacy K. Conrad*



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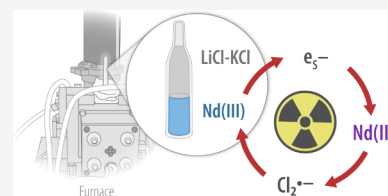
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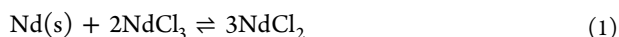
ABSTRACT: In the pyrochemical reprocessing of used nuclear fuel, understanding the effects of ionizing radiation on the chemical behavior of fission products in molten chloride eutectic salt is crucial. This study investigates the impact of radiation on the chemical and kinetic speciation of Nd ions in molten LiCl-KCl eutectic, which are challenging to separate due to their multivalent oxidation states and comproportionation reactions. Electron pulse radiolysis techniques were used to determine transient absorption spectra, chemical kinetics, and activation parameters for the reaction of Nd ions with the primary products of molten LiCl-KCl eutectic radiolysis: the solvated electron (e_s^-) and dichlorine radical anion ($Cl_2^{\bullet-}$).

Initially, Nd(III) reacted rapidly with e_s^- , forming Nd(II) with a second-order rate coefficient of $k = (4.54 \pm 0.07) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ at 673 K. The resulting Nd(II) ions then reacted slowly with $Cl_2^{\bullet-}$, regenerating Nd(III) with an estimated rate of $k = (1.72 \pm 0.04) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ at 673 K. Additionally, the comproportionation reaction of Nd(III) and Nd(s) to form Nd(II) was monitored chemically and electrochemically. This work suggests that the comproportionation reaction of Nd(III) with Nd(s) to form Nd(II) is unstable in molten LiCl-KCl eutectic.



1. INTRODUCTION

Understanding the electrochemically-induced transitions of rare earth elements is essential for developing innovative recovery methods for recycling and separating end-of-life products from energy sources, such as in the recovery of permanent magnets or in the pyrochemical reprocessing of used nuclear fuel (UNF).^{1,2} For instance, in pyrochemical reprocessing, metallic UNF is dissolved in molten lithium chloride-potassium chloride (LiCl-KCl) eutectic salt at 773 K. Uranium metal can then be recovered onto an immersed cathode^{3,4} via electrorefining for fabrication into new fuel. As the uranium is recovered in this manner, the remaining fission products, including rare earth elements, accumulate in the molten salt. To preserve process efficiency and maximize salt recyclability, fission products, specifically rare earth elements, can be removed by electrodeposition onto a liquid metal^{5,6} or a solid inert cathode. The latter process, known as lanthanide/actinide drawdown,⁷ has a relatively low recovery efficiency for certain rare-earth elements, such as neodymium (Nd): 30–55%.⁸ Nd is an abundant rare earth fission product in UNF, accounting for ~7% of the overall rare earth fission product yield.⁹ The reduced recovery efficiency of Nd metal is due to its ability to access multiple oxidation states and undergo a comproportionation reaction in molten chloride salt:⁵



Due to the equilibrium nature of this reaction, comproportionation can also reverse to disproportionation. The

occurrence of either reaction depends on the stability and temperature of the molten chloride salt. Additionally, rare earth metals can undergo dissolution in molten LiCl-KCl eutectic when exposed to the melt for extended periods of time.^{10,11} These complicating factors can alter the melting point, viscosity, and ionic conductivity of the melt, which affects the longevity and recyclability of the recovery process.¹² While the two-step reduction for Nd(III)/Nd(s) has been measured using multiple electroanalytical techniques,^{13,14} the stability of Nd(II) in molten chloride salt and the effects of Nd(II) on the recovery efficiency of Nd metal are not well understood. Furthermore, the impacts of ionizing radiation on this system and the final redox distribution of Nd remain unknown.

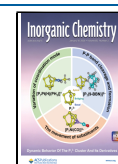
Given previous findings on the radiation-induced chemical behavior of zinc (Zn) and chromium (Cr) ions in molten LiCl-KCl eutectic,^{15,16} the redox distribution of Nd(II,III) is expected to be strongly dependent on their respective interactions with the primary products of salt radiolysis. Initially, the absorption of ionizing radiation by a simple

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molten chloride salt such as LiCl-KCl eutectic, results in the formation of solvated electrons (e_s^-) and chlorine atoms ($Cl^\bullet$):^{15–17}



which can combine with a neighboring Cl^- to form a dichlorine radical anion ($Cl_2^{\bullet-}$):



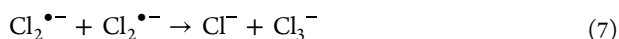
In a perfectly pure salt, the e_s^- reacts with either $Cl^\bullet$ or $Cl_2^{\bullet-}$ to reform Cl^- :



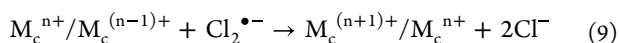
and the excess energy is dissipated as heat into the surrounding medium. However, in real-world salts, trace metal ion contaminants (M_c^{n+}) are inevitable and have an affinity for reacting with the e_s^- , creating a break in the charge recombination process:



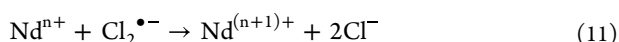
Under these conditions, the $Cl_2^{\bullet-}$ is susceptible to disproportionation:



and reactions involving $M_c^{n+}/M_c^{(n-1)+}$ species, as demonstrated previously for Zn(II) and Cr(II,III) ions:^{15,16}



In the case of Nd(II,III), the following processes are plausible, albeit mediated by competition kinetics:



This dynamic scenario can be further complicated by the formation of metal particulates seeded by radiolytically reduced species such as Nd^+ and Nd^0 . Similar behavior has been observed for nickel (Ni) ions in molten $ZnCl_2$ exposed to X-rays.¹⁸ Consequently, accurately predicting the redox distribution of Nd in irradiated molten chloride salts and determining the impacts of radiolysis on the efficiency of pyrochemical processes using liquid metal^{5,19} and inert cathodes,⁷ requires a comprehensive understanding of the chemical kinetics underpinning eq 10 and 11.

To address this knowledge gap, this work focused on the radiation-induced chemistry of Nd(II,III) ions in molten LiCl-KCl eutectic, applying electron pulse radiolysis techniques to determine transient absorption spectra, chemical kinetics, and activation parameters (Arrhenius and Eyring) for eq 10 and 11. These time-resolved measurements were complemented by electrochemical and optical spectroscopy techniques, which were employed to identify the characteristic absorption bands of Nd(III) and Nd(II) in molten LiCl-KCl eutectic.

2. METHODS

2.1. Materials

Acetone (99.9% reagent grade), ethanol (absolute anhydrous), isopropanol (IPA, 99.9% reagent grade), lithium chloride-potassium

chloride eutectic (LiCl-KCl, 99.99% trace metal basis anhydrous beads), potassium thiocyanate (KSCN, $\geq 99.0\%$ ACS reagent grade), and silver chloride (AgCl, 99.999% anhydrous) were procured from MilliporeSigma. Neodymium chloride ($NdCl_3$, anhydrous, 99.995% trace metal basis), silver wires (Ag, 99.99%, 1 mm diameter), and tungsten wires (99.9%, 1 mm diameter and 15 cm length) were purchased from Thermo Fisher Scientific. A neodymium metal rod (99.9%, 3 mm diameter and 15 cm length) was sourced from Stanford Instruments. Argon ($>99.998\%$) and nitrous oxide (UHP) gases were supplied by Airgas.

2.2. Sample and Electrode Preparation

All salt manipulations were performed in an argon atmosphere (<0.1 ppm of O_2 , <0.1 ppm of H_2O) glovebox (VTI, VTISS1–1809–0100) at Idaho National Laboratory (INL).

2.2.1. Sample Salt Mixtures. Mixtures of 36 and 65 mM of $NdCl_3$ in LiCl-KCl eutectic were prepared by weighing the desired amounts of $NdCl_3$ and LiCl-KCl eutectic salt using a Mettler Toledo balance (Model: TLE204E). The resulting salt mixtures were loaded into a Sigradur glassy carbon tapered crucible (GAT6, 64 mL HTW Germany), heated to 373 K for 2 h, and then to 543 K inside a Fisher Scientific Thermolyne box furnace (Model: FB1315M) for 2 h to remove any residual moisture. The salt mixtures were then melted at 973 K and maintained at that temperature for 3 h to ensure homogeneity before cooling to room temperature (~ 298 K) and grinding into a fine powder using an agate mortar and pestle.

2.2.2. Pulse Radiolysis Samples. A stock mixture of 130 mM $NdCl_3$ in LiCl-KCl eutectic was prepared and samples for irradiation were made by diluting the stock salt mixture with additional LiCl-KCl eutectic, to provide a Nd(III) concentration range of 0–50 mM (wt % compositions are provided in Table S1 of the Supporting Information, SI). Once prepared, the sample salt mixtures were crushed and loaded into 5×5 mm quartz cuvettes (Spectrocell, RF-305) custom-made with elongated stems and Teflon valves (Chemglass, CG-512). Prior to salt loading, the cuvettes were cleaned with IPA and baked under argon at 873 K overnight. Once loaded, the cuvettes were sealed under argon using the valves and transferred to a bespoke Schlenk line where they were flame-sealed under slight vacuum (~ 13 kPa). The samples were then securely shipped to Brookhaven National Laboratory (BNL) for pulsed electron irradiation.

2.2.3. Electrodes. The surface of the tungsten wires of the working and counter electrodes (WEs and CE, respectively) were polished with silicon carbide sanding paper (up to 800 grit) and then cleaned with IPA. Afterward, both electrodes were inserted into a two-hole alumina tube (McMaster Carr, 1.02 mm inner diameter, 3 mm outer diameter) and placed in the cell. To fabricate the reference electrode (RE), the appropriate amounts of AgCl and LiCl-KCl eutectic were weighed to give a 1 wt % AgCl-LiCl-KCl salt mixture, which was then premelted, crushed and poured into a Ceramic Solutions mullite tube (6.35 mm outer diameter, 4.77 mm inner diameter, and 30.4 cm length). Prior to salt loading, the bottom of the mullite tube was sanded to provide better connectivity with the salt. A silver wire was polished by the same procedure used for the WEs and CEs and then placed inside the salt-loaded mullite tube.

2.3. Electron Pulse Radiolysis Experiments

Chemical kinetics for the reaction of Nd(III) with the e_s^- and $Cl_2^{\bullet-}$ in molten LiCl-KCl eutectic were measured at BNL using the Laser Electron Accelerator Facility (LEAF)²⁰ and colocated Van de Graaff (VdG) electron accelerator.²¹ Individual flame-sealed salt samples (Table S2) were loaded into bespoke high temperature-controlled cuvette holders^{15,22} installed at the end of the LEAF and VdG accelerator beamlines. Prior to making a measurement, each loaded sample was heated above their melting temperature (>623 K) and left for 20 min to reach thermal equilibrium. Sample temperatures were measured with a K-type thermocouple installed ~ 1 cm from the sample in each cuvette holder.

2.3.1. LEAF Irradiations. The LEAF system was used to measure transient absorption spectra (300–1050 nm) and reaction kinetics over the initial 20 μs following a ~ 30 ps pulse of 8.7 MeV electrons.

Transient spectra measurements involved averaging 3–4 electron pulses per wavelength to minimize the accumulated dose to the salt. Kinetic measurements comprised of 4–8 individual pulses in triplicate, which were then averaged. Wavelength selection was achieved using a series of Edmund Optics bandpass interference filters (fwhm: ~ 10 nm < 800 nm, and fwhm: ~ 25 nm ≥ 800 nm). Signals were captured at a given wavelength as a function of time using either a silicon PIN photodiode (FND-100QH, EG&G Optoelectronics) or a silicon PIN photodiode (SS973–01, Hamamatsu Photonics).²⁰ Collected signals were digitized using a Teledyne LeCroy WaveRunner oscilloscope, either HRO 66Zi (600 MHz, 12 bit) or 640Zi (4 GHz, 8 bit). The choice of detector for a given measurement was based upon the lifetime of the targeted transient.

2.3.2. VdG Irradiations. To access higher doses per pulse for elucidating second-order processes, the VdG electron accelerator was used. Irradiations consisted of 0.1–1.0 μ s duration pulses of 2.0 MeV electrons. A monochromator (Gemini 180, Horiba) with a set of long-pass filters (200, 250, 280, 385, 475, and 630 nm) was used for wavelength selection. Optical signals were recorded using a photomultiplier tube (9659QB, EMI Electronics Ltd.) and a LeCroy HDO 4034 oscilloscope (350 MHz, 12 bit).

2.3.3. Dosimetry. Electron pulse dosimetry was performed each day at LEAF under ambient temperature conditions using the decay of the hydrated electron in aqueous solution at 650 nm ($G_e = 7.2 \times 10^{-4}$ m² J⁻¹) for up to 5 μ s.²³ For the VdG, dosimetry was measured by monitoring the growth and decay of the thiocyanate radical anion dimer, (SCN)₂^{•-} from irradiated N₂O-saturated 10 mM KSCN solutions monitored at 475 nm ($G_e = 2.59 \times 10^{-4}$ m² J⁻¹).²⁴ The resulting absorbed doses were subsequently adjusted to account for the electron density difference between the ambient temperature (~ 298 K) aqueous dosimeter solution and the molten LiCl-KCl eutectic base-salt at a given temperature. These doses were further adjusted by a factor of 0.86 to correct for the mass attenuation difference between the aqueous dosimeters and the molten salt mixtures. This correction factor was determined by calculating the number of electrons per unit mass in NdCl₃-LiCl-KCl, considering the mass fractions of each component. Given the low mass fraction of NdCl₃ in the samples, the dose ratio for the molten salts was approximated using the values for neat LiCl-KCl eutectic. The final doses per LEAF or VdG electron pulse are summarized in Table S3.

2.3.4. Absorption Spectra Kinetic Analysis (SK-Ana). The time-resolved absorption spectra collected at both accelerators comprised of multiple absorbing and often overlapping species: the e_s⁻, Cl₂^{•-}, and Nd(II). These components and their respective kinetics were deconvoluted using SK-Ana software (v. 3.4, Pernot, 2018),²⁵ which utilizes singular value decomposition and alternating least squared methods.

2.4. Comproportionation Measurements

To achieve chemical comproportionation, a 3 mm diameter Nd metal rod was submerged in a molten 36 mM NdCl₃-LiCl-KCl salt mixture at 773 K. The electrolyte mixture was initially loaded into a Spectrocell quartz cuvette (20 mm $\times$ 10 mm) precleaned with IPA and dried under vacuum at 398 K for 3 h. Approximately 10 g of the salt mixture was poured into the cuvette and heated above its melting point to 773 K. Once the salt was molten, the Nd rod was submerged in the salt mixture, and the system was monitored continuously for 30 min until the Nd rod was removed. Optical absorption spectra were obtained using an Ocean Optics HR6 high-resolution diode array spectrometer, which was equipped with a DH-2000-BAL deuterium/tungsten light source (covering 250–950 nm). Each measurement involved a 30 s acquisition time, with two scans averaged per spectrum. The data were processed using SpectraGryph 1.2 software,²⁶ with baseline corrections applied utilizing adaptive and scattering functions. The raw data, experiment photo conditions, along with the baseline correction instructions for the spectroscopy figures can be found in the SI under Kinetic Parameters and Spectroscopy Raw Data.

2.5. Spectro-electrochemical Measurements

Further insight into the chemical behavior and spectral properties of Nd(II,III) ions in molten LiCl-KCl eutectic salt mixtures was obtained using electrochemical methods. A 65 mM NdCl₃-LiCl-KCl salt mixture was loaded into a Spectrocell quartz cuvette (20 mm $\times$ 10 mm) and then placed in a stainless-steel rack. The entire spectro-electrochemical cell was then positioned inside a furnace and heated to 773 K. Under these conditions, the RE was immersed and allowed to equilibrate overnight. After approximately 18 h, cyclic voltammetry and chronoamperometry techniques were applied to the salt mixture to investigate the electrochemical behavior of Nd(III). The WE was immersed in the molten salt for 30 min before initiating any electrochemical measurements. The WE, CE, and RE were arranged inside the cuvette as shown in Figure 1. Electrochemical measure-



Figure 1. Three-electrode cell employed for electrochemical measurements of molten NdCl₃-LiCl-KCl eutectic at 773 K.

ments were performed using a Gamry 1010E potentiostat-galvanostat, with cell temperatures monitored by a K-type thermocouple. To detect the formation of Nd(II), simultaneous in situ UV–vis absorption spectroscopy measurements were taken during electrolysis using an Ocean Optics HR6 high-resolution diode array spectrometer with a DH-2000-BAL deuterium/tungsten light source (covering 250–1100 nm) with a 2–4 min acquisition time, an integration time between 0.04–0.06 s, and two scans averaged per spectrum. The data were processed using SpectraGryph 1.2 software, with baseline corrections as detailed in SI. An internal resistance (IR) correction was applied to the cyclic voltammogram by performing potentiostatic electrochemical impedance spectroscopy (EIS) to determine the solution resistance, followed by manual corrections using Ohm's law.

2.6. Characterization of Molten Chloride Products

Compositional analysis of the NdCl₃-LiCl-KCl salt mixtures used in this study, before and after irradiation or electrochemical manipulation, was determined with an accuracy of $< 5\%$ by inductively coupled plasma atomic emission spectroscopy (ICP-AES) using a Thermo Fisher iCap Duo. A summary of these values is provided in Table S1.

3. RESULTS AND DISCUSSION

3.1. Transient Spectra in Molten NdCl₃-LiCl-KCl Salt Mixtures

The radiation-induced absorption spectra for molten LiCl-KCl eutectic and 10 mM NdCl₃-LiCl-KCl at selected times following an electron pulse at 673 K are shown in Figure 2a,b, respectively. For the neat molten LiCl-KCl eutectic, two transient species were observed which are typical for this molten salt system. The deconvoluted spectra and species

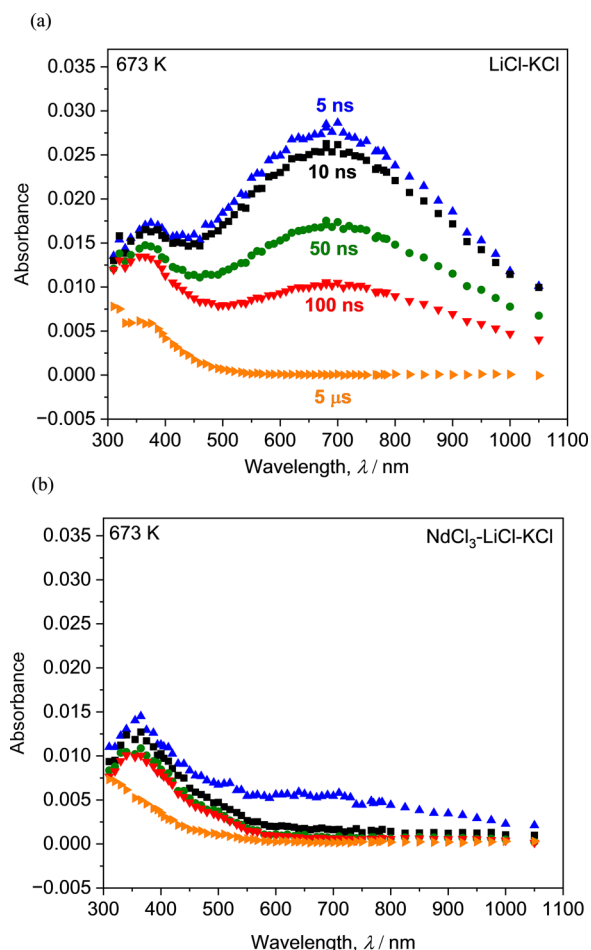


Figure 2. Transient absorption spectra for molten (a) LiCl-KCl eutectic and (b) 10 mM NdCl₃-LiCl-KCl at select times following an electron pulse at 673 K.

kinetics for neat LiCl-KCl are shown in SI Figure S1. These species were assigned to the e_s^- ($\lambda_{\max} \sim 700$ nm) and $Cl_2^{\bullet-}$ ($\lambda_{\max} \sim 350$ nm), in agreement with previous studies.^{15,17} Here, the e_s^- decayed within 600 ns (short-lived), whereas $Cl_2^{\bullet-}$ remained for over 1 μ s (longer-lived). When 10 mM NdCl₃ was added to the salt the e_s^- absorption band was suppressed (Figure 2b), indicating that Nd(III) is an effective scavenger of the e_s^- .

In the presence of 10 mM NdCl₃, the transient spectra contained three statistically consistent principal components, including a new peak at 500 nm, as shown in Figure 2b. To deconvolute these spectra with SK-Ana, the isolated absorption spectrum of $Cl_2^{\bullet-}$ from neat LiCl-KCl eutectic (Figure S1) was fixed, while the other two components were derived. The rotational ambiguity plots for these fits are shown in Figure S2. The deconvoluted spectra (Figure 3a) show a mixture of the e_s^- , $Cl_2^{\bullet-}$, and a species growing in the visible region, which appears as a shoulder on the $Cl_2^{\bullet-}$ absorption band in Figure 2b.

This component was found to have a deconvoluted $\lambda_{\max}$ at approximately 500 nm, which is presumed to be Nd(II) based on previous molten salt spectroscopy of NdCl₂ by Hayashi et al., who observed a peak evolving at 500 nm when Nd metal was added to molten NdCl₃-LiCl-KCl.²⁷ The e_s^- was again observed with an $\lambda_{\max}$ of approximately 700 nm (Figure 3a); however, its absorption intensity and lifetime were significantly

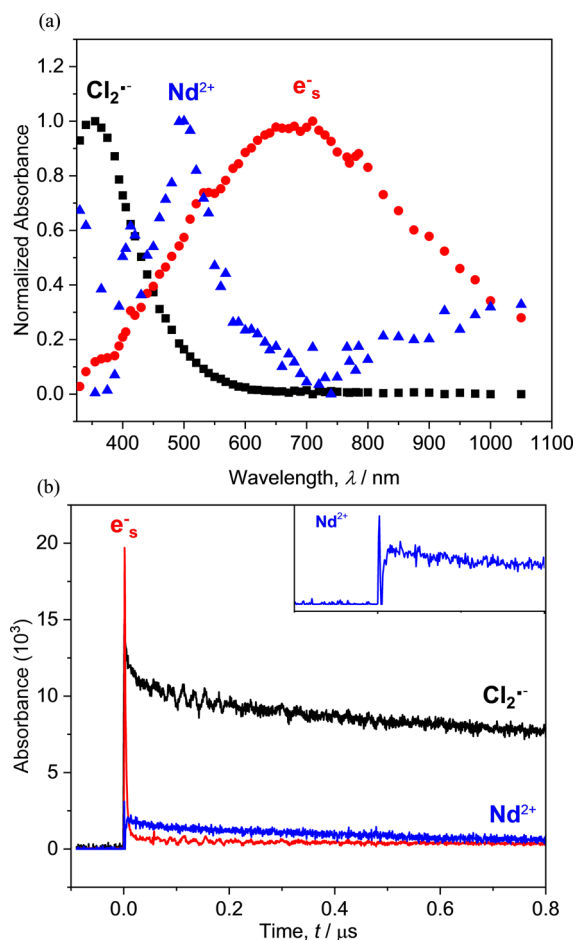


Figure 3. (a) Normalized spectra of components from SK-Ana deconvolution of molten 10 mM NdCl₃-LiCl-KCl up to 1 μ s following an electron pulse at 673 K. (b) SK-Ana derived kinetics at the wavelength absorption maxima for each component: the e_s^- , $Cl_2^{\bullet-}$, and Nd(II).

reduced compared to neat LiCl-KCl eutectic, with no e_s^- remaining after 10 ns. These optical changes are consistent with chemical reactivity between Nd(III) and the e_s^- (eq 10). Although the $Cl_2^{\bullet-}$ spectrum was fixed for deconvolution purposes, analysis of its derived kinetics (Figure 3b) indicates that it is formed with the electron and then slowly decays. The additional third component, was found to have a rate of formation equivalent to the decay of the e_s^- (Figure 3b), further supporting its assignment as Nd(II), produced from eq 10.

To confirm the growth and decay of Nd(II) in the irradiated NdCl₃-LiCl-KCl salt mixtures, a series of spectroscopic and spectro-electrochemical measurements were made using a high-temperature spectroscopy furnace²² inside an inert atmosphere glovebox. First, Nd(II) formation was chemically induced via comproportionation (eq 1), wherein a Nd metal rod was submerged in molten 36 mM NdCl₃-LiCl-KCl at 773 K. The resulting absorption spectra are shown in Figure 4. Initially, the spectrum consisted of the absorption bands for Nd(III) only. Once the Nd metal rod was immersed in the salt mixture, a new absorption band developed over time at approximately 430–500 nm (Figure S3), and all the Nd(III) bands decreased slightly. After 30 min, the accumulation of Nd(II) darkened the molten salt, increasing the background (Figure S3). Yamana et al. determined that the Nd(II)

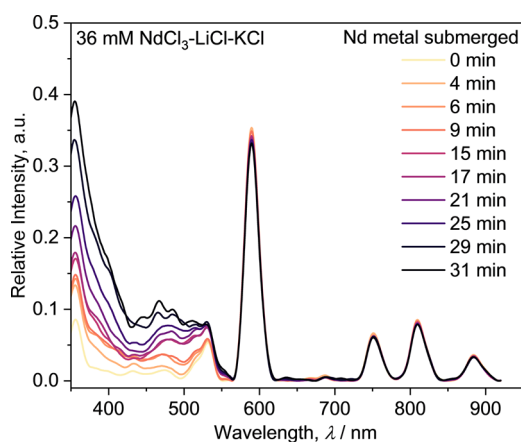


Figure 4. Absorption spectra for molten 36 mM $\text{NdCl}_3\text{-LiCl-KCl}$ at select times after submerging a neodymium metal rod at 773 K.

absorption band represented an $f-d$ transition in molten LiCl-KCl eutectic,^{27,28} indicating that this transition leads to a high molar absorptivity. In addition to salt darkening due to the formation of Nd(II) , the quartz cuvette was etched. Etching contributed to light scattering, further modifying the background. These observations align with the findings of Hayashi et al., who attributed a broad peak at ~ 500 nm to Nd(II) and observed an increase in the spectrum baseline due to the darkening of the salt and quartz etching. Depending on the salt conditions, Nd(II) can disproportionate into Nd metal and Nd(III) (eq 1). The Nd metal formed in this way can also react with the quartz cuvette. Yamana et al. observed similar behavior, noting that Nd(II) reacts with the quartz cuvette, leading to light scattering, and that Nd metal interacts with the quartz surface.

Further confirmation of the absorption spectrum of Nd(II) and the redox behavior of this species in molten LiCl-KCl eutectic was attained through a combination of electrochemical techniques. First, the electrochemical behavior of Nd(II,III) ions in molten LiCl-KCl was investigated using cyclic voltammetry at 773 K (Figure 5). The initial cathodic reduction at -1.8 V corresponds to the reduction of Nd(III) to Nd(II) , followed by the reduction of Nd(II) to Nd(s) at -2.0 V, indicating a multistep electron transfer. The observed

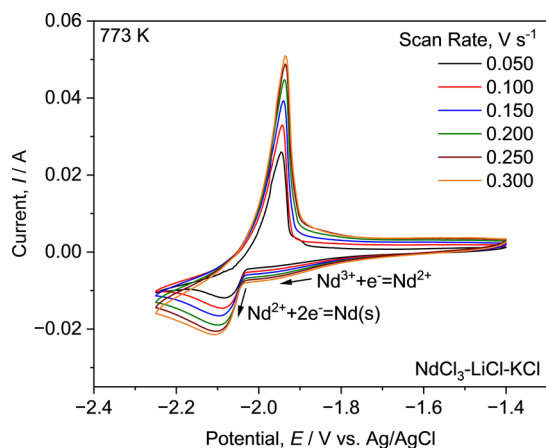


Figure 5. Cyclic voltammograms of 65 mM $\text{NdCl}_3\text{-LiCl-KCl}$ at 773 K using tungsten WE vs Ag/AgCl over a scan rate range of 0.050–0.300 V s^{-1} . Surface area of WE: 0.134 cm^2 , IR corrected.

double reduction–oxidation waves remained consistent between different scan rates ($0.050\text{--}0.300 \text{ V s}^{-1}$), consistent with previous studies.^{13,14,29}

Chronoamperometry was then employed to reduce Nd(III) to Nd(II) and observe any change in its absorption spectra. A potential of -1.95 V (vs Ag/AgCl) was applied for 24 min, as shown in Figure 6a. As the solution darkened, an increase in

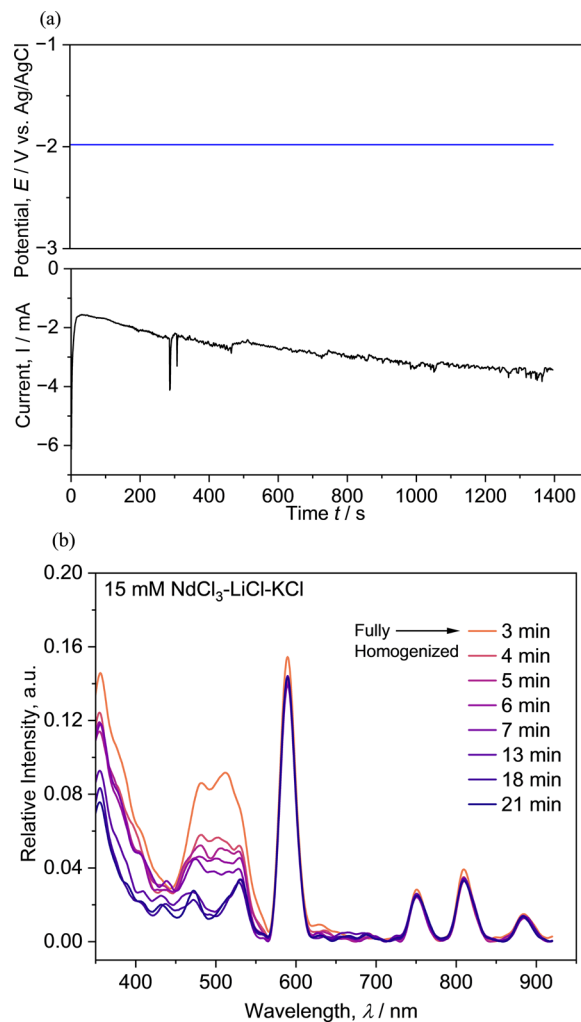


Figure 6. (a) Current response vs applied potential (-1.95 V vs Ag/AgCl) and (b) Absorption spectra for molten LiCl-KCl eutectic at select times following the addition of ~ 2 g of the electrochemically induced salt (65 mM $\text{NdCl}_3\text{-LiCl-KCl}$) at 773 K.

the cathodic current was detected, but no significant growth of the Nd(II) absorption band at ~ 500 nm was observed (Figure S4). After 24 min, the applied potential was stopped, the salt mixture was recovered, and a sample was taken to dilute in LiCl-KCl eutectic with a final concentration of ~ 15 mM of Nd(III,II) at 773 K. The UV–vis spectra following the dissolution of the recovered salt, Figure 6b, revealed that the Nd(II) absorption band decreased as the solution reached equilibrium. Under the investigated conditions, Nd(II) disproportionation was complete within 10 min, as evidenced by the remaining absorption bands belonging to Nd(III) and the appearance of suspended metallic Nd particles (Figure S5). These observations indicate that Nd(II) is not stable at high temperatures in molten LiCl-KCl eutectic.

Comparison of the chemically and electrochemically induced Nd(II) spectra (Figure 6b) with the third component from SK-Ana deconvolution in the pulse radiolysis experiments (Figure 3a) confirms our assignment of Nd(II). Further, the observed rate of Nd(II) disproportionation (minutes) indicates that the decay of the Nd(II) signal in Figure 3b (seconds) cannot be solely attributed to eq 1 in radiation environments.

3.2. Solvated Electron Reduction of Nd(III)

To determine the rate coefficient for eq 10, the decay of the e_s^- over a range of Nd(III) concentrations (1–10 mM NdCl_3) was fitted using a single exponential decay beginning at 0.4 ns after the electron pulse to obtain pseudo first-order rate coefficients (k' , s^{-1}) at a given temperature (673–873 K). These fits are given in Figure 7a for 673 K, and Figure S6 for the remaining temperatures.

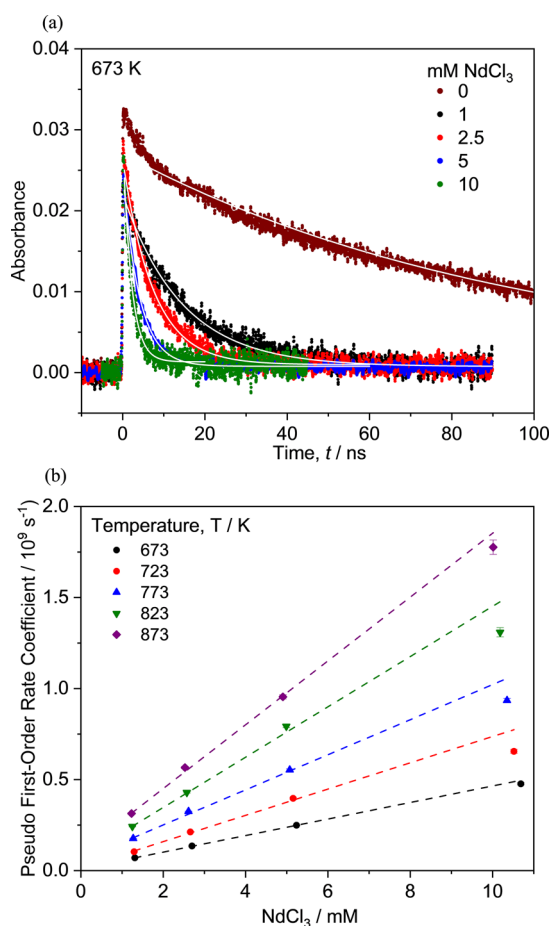


Figure 7. (a) Fitted 671 nm absorbances for the decay of the e_s^- in molten LiCl-KCl and 1–10 mM NdCl_3 -LiCl-KCl following an electron pulse at 673 K. (b) Linear fits used for deriving second-order rate coefficients for the reaction of Nd(III) with the e_s^- for each investigated temperature. Individual data points are the fitted pseudo first-order decay rate coefficients.

Linear fits to these pseudo first-order rate coefficients plotted against the density-corrected Nd(III) concentration gave the corresponding second-order rate coefficients (k , $\text{M}^{-1} \text{s}^{-1}$) for each investigated temperature, which conform to Arrhenius behavior, i.e., increasing rate with increasing temperature. This process is presented in Figure 7b, and the derived k -values are summarized in Table 1.

Table 1. Second-Order Rate Coefficients (k) Calculated for the Reaction of Nd(III) with the e_s^- in Molten LiCl-KCl Eutectic

Temperature (K)	Second-order rate coefficient ($\times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$)
673	4.54 ± 0.07
723	7.21 ± 0.66
773	9.63 ± 0.85
823	13.80 ± 0.85
873	17.53 ± 0.68

By plotting the natural logarithm of each temperature-dependent k -value as a function of the inverse temperature ($1/T$), the Arrhenius parameters can be derived for eq 10, as shown in Figure 8. This gave an activation energy (E_a) of 33.20 ± 0.42

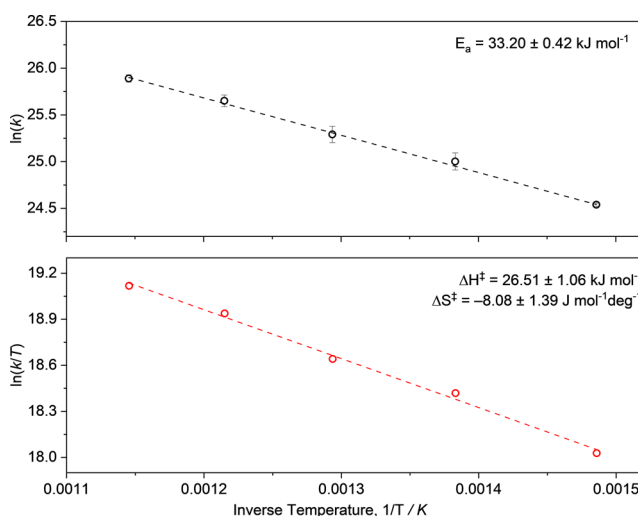


Figure 8. Arrhenius and Eyring plots for the reaction of Nd(III) with the e_s^- in molten LiCl-KCl eutectic.

kJ mol^{-1} , and a pre-exponential factor (A) of $1.71 \pm 0.13 \times 10^{13} \text{ M}^{-1} \text{ s}^{-1}$. These values are comparable with previous measurements for the reduction of Cr(III) by the e_s^- in molten LiCl-KCl eutectic: $E_a = 33.50 \pm 0.6 \text{ kJ mol}^{-1}$ and $A = 1.70 \pm 0.2 \times 10^{13} \text{ M}^{-1} \text{ s}^{-1}$.^{15,16}

Additionally, plotting the natural logarithm of k/T against $1/T$ yields the corresponding Eyring activation parameters for eq 10: an enthalpy of activation ($\Delta H^\ddagger$) of $26.51 \pm 1.06 \text{ kJ mol}^{-1}$, and an entropy of activation ($\Delta S^\ddagger$) of $-8.08 \pm 1.39 \text{ J mol}^{-1} \text{ deg}^{-1}$. The negative $\Delta S^\ddagger$ value indicates that the reduction of Nd(III) to Nd(II) proceeds through a single transition state species via an associative reaction mechanism. Moreover, the e_s^- acts a “structure-breaking” ion in polar media³⁰ which increases a system’s entropy. When the e_s^- is removed by reacting with Nd(III), the extended networks of the salt can reestablish,^{31–33} leading to a decrease in the system’s entropy.

3.3. Dichlorine Radical Anion Oxidation of Nd(II)

As discussed in the introduction, $\text{Cl}_2^{\bullet-}$ can react with metal ion solutes and impurities (eq 9) in competition with charge recombination (eq 5) and disproportion (eq 7) processes. The rate of $\text{Cl}_2^{\bullet-}$ decay changed in the presence of NdCl_3 , as shown in Figure 9a. However, this change was independent of the concentration of NdCl_3 (1–10 mM) at a given applied dose, as summarized in Figure S7. These observations indicate that Nd(III) is not oxidized to Nd(IV) by $\text{Cl}_2^{\bullet-}$ (eq 11).

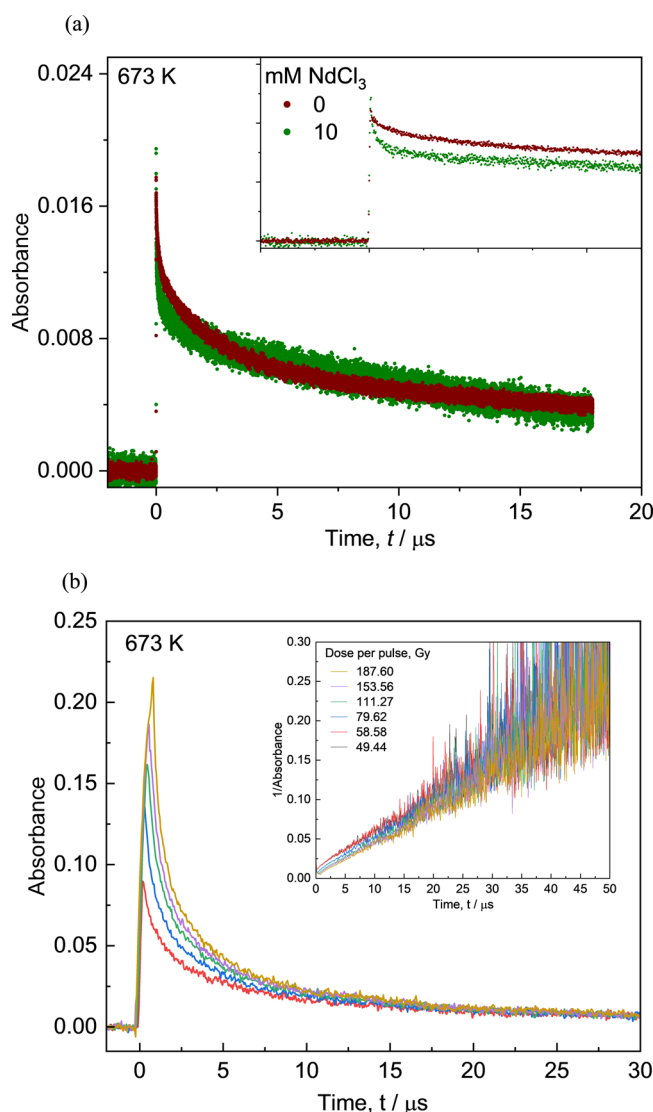


Figure 9. (a) Decay kinetics for $\text{Cl}_2^{\bullet-}$ at 340 nm for molten LiCl-KCl and 10 mM NdCl_3 -LiCl-KCl and (b) transient absorbances for the decay of $\text{Cl}_2^{\bullet-}$ at 340 nm in molten 50 mM NdCl_3 -LiCl-KCl eutectic as a function of dose per electron pulse at 673 K. Inset: the corresponding reciprocal absorbance plot.

Instead, the increased rate of $\text{Cl}_2^{\bullet-}$ decay is due to its reaction with the Nd(II) product from eq 10 via eq 11.

Assuming the reaction occurs between two radiolysis products, eq 11 should obey second-order kinetics, and the rate should remain constant with respect to the applied dose, as the two reactants are generated in approximately equal amounts. The corresponding reciprocal plot of the absorbance at 340 nm versus time is given in Figure 9b for the VdG irradiation of 50 mM NdCl_3 -LiCl-KCl at 673 K.

Figure 9b clearly demonstrates that the reciprocal absorbance plots for the decay of $\text{Cl}_2^{\bullet-}$ exhibit linear behavior and are independent of the applied dose, confirming that the reaction of $\text{Cl}_2^{\bullet-}$ with Nd(II) via eq 11 follows second-order kinetics. Since this relationship remains consistently linear over time, the results indicate that this reaction is the dominant $\text{Cl}_2^{\bullet-}$ decay pathway, out-competing any contributions from eq 7. The corresponding rate coefficients ($k/\epsilon l$) as a function of the dose per pulse are summarized in Figure S8. At the significantly lower doses applied using LEAF (3.9 Gy pulse⁻¹),

the apparent rate for eq 11 is higher because the relatively low yield of $\text{Cl}_2^{\bullet-}$ per pulse means that eq 11 is now in competition with reactions of Nd(II) and metal impurities (eq 9) in the system. In contrast, the higher doses applied using the VdG (47–189 Gy pulse⁻¹) produce enough Nd(II) and $\text{Cl}_2^{\bullet-}$ to outcompete other potential decay pathways. Linear fitting of the reciprocal absorbance plots yields an average second-order rate coefficient of $k_{673\text{ K}} = (1.72 \pm 0.01) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$, where the path length is 0.5 cm, and the $\epsilon = 8000 \text{ M}^{-1} \text{ cm}^{-1}$. The ϵ -value was derived from adjusting the value from $\text{Cl}_2^{\bullet-}$ in aqueous solution according to Hug³⁴ based on the work by Makarov et al.³⁵ The rate coefficient obtained from the reciprocal absorbance plots aligns well with previously reported rate coefficients obtained for the reactions of Zn(I) and Cr(II) with $\text{Cl}_2^{\bullet-}$: $k_{\text{Zn(I)}} = 2.2 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ and $k_{\text{Cr(II)}} = 1.70 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$.

4. CONCLUSIONS

This study investigated the chemical kinetics underlying the reactions of Nd(III,II) with transient radical products (e_s^- and $\text{Cl}_2^{\bullet-}$) arising from the electron pulse irradiation of molten LiCl-KCl eutectic. The e_s^- was found to rapidly reduce Nd(III) to Nd(II) at a rate of $k = (4.54 \pm 0.07) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ at 673 K, with the concomitant formation of Nd(II) . The formation of Nd(II) was confirmed through complementary electrochemical and optical spectroscopy methods, which showed that Nd(II) exhibits an absorption band in the wavelength range of 430–500 nm, consistent with the transient growth observed in the pulsed electron radiolysis spectra. The Nd(II) formed through radiolysis was found to be shorter lived compared to its chemically or electrochemically generated counterpart (microseconds vs seconds), which is inherently unstable and prone to disproportionation. This is a consequence of Nd(II) , and not Nd(III) , reacting with $\text{Cl}_2^{\bullet-}$ at a rate of $k = (1.72 \pm 0.01) \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ at 673 K. This reoxidation process limits further decomposition of the salt.

The instability of Nd(II) at high temperatures and its rapid cycling between oxidation states complicate the recovery of Nd metal in pyrochemical reprocessing of UNF. These findings enhance our understanding of this complex redox behavior in molten chloride salts under both chemical and radiolytic conditions, providing valuable insights for developing more efficient pyrochemical reprocessing techniques.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.5c04788>.

Chemical analysis of all the salt compositions (in wt.% and mM) containing NdCl_3 in LiCl-KCl eutectic salt; kinetic parameters and absorption spectra of electron pulse irradiation of molten LiCl-KCl eutectic and NdCl_3 containing salts; raw data for the absorption spectra from the figures described in the manuscript, along with an explanation of Spectragryph parameters (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Stephanie Castro Baldivieso – Center for Radiation Chemistry Research and Glenn T. Seaborg Institute, Idaho National Laboratory, Idaho Falls, Idaho 83415, United

States; orcid.org/0000-0003-3610-1762;

Email: stephanie.castrobaldivieso@inl.gov

Jacy K. Conrad – Center for Radiation Chemistry Research, Idaho National Laboratory, Idaho Falls, Idaho 83415, United States; orcid.org/0000-0002-0745-588X; Email: jacy.conrad@inl.gov

Authors

Gregory P. Horne – Center for Radiation Chemistry Research, Idaho National Laboratory, Idaho Falls, Idaho 83415, United States; orcid.org/0000-0003-0596-0660

Davis Bryars – Department of Nuclear Engineering, North Carolina State University, Raleigh, North Carolina 27607, United States

Alejandro Ramos Ballesteros – Center for Radiation Chemistry Research, Idaho National Laboratory, Idaho Falls, Idaho 83415, United States; orcid.org/0000-0002-4350-6824

Andrew R. Cook – Department of Chemistry, Brookhaven National Laboratory, Upton, New York 11973, United States; orcid.org/0000-0001-6633-3447

Bobby Layne – Department of Chemistry, Brookhaven National Laboratory, Upton, New York 11973, United States

Kazuhiro Iwamatsu – Department of Chemistry, Brookhaven National Laboratory, Upton, New York 11973, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.inorgchem.5c04788>

Author Contributions

Baldivieso: conceptualization, methodology, validation, formal analysis, investigation, data curation, writing (original draft), writing (review and editing), visualization, and funding acquisition; Horne: conceptualization, methodology, formal analysis, data curation, writing (original draft), writing (review & editing), visualization, supervision, and funding acquisition; Bryars: investigation, data curation, and writing (review and editing); Cook: methodology, validation, formal analysis, and writing (review and editing); Layne: methodology and writing (review and editing); Ramos Ballesteros: methodology, validation, data curation, and writing (review and editing); Iwamatsu: methodology, formal analysis, and writing (review and editing); Conrad: methodology, formal analysis, data curation, writing (original draft), writing (review and editing), visualization, and supervision.

Notes

The authors declare no competing financial interest.

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